# Principles of Exercise in Cardiorespiratory Care

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# Learning Outcomes

- Understanding the cardiorespiratory benefits of physical activity and exercise
- Understanding the current recommended guidelines for physical activity
- Able to appropriately prescribe suitable exercise for targeted cardiorespiratory population
- Understand how to measure and test exercise capacity
- Understand the current state of pulmonary rehabilitation

# What is exercise?

# What is physical activity?

- **Physical Activity**
  - According to the World Health Organisation (WHO), physical activity is any bodily movement produced by skeletal muscles that requires energy expenditure
  - Physical activity can be categorised into occupational, sports, conditioning, and household

- **Exercise**
  - The term "exercise", is a subcategory of PA.
  - Planned, structured, repetitive form of activity that aims to improve or maintain one or more components of physical fitness
  - For conditioning and improving the function of any part of the body

- **Physical rehabilitation**
  - Physical rehabilitation aims to restore movement and physical function towards fitness. .
  - Dietz Model of rehab
    - preventative
    - restorative
    - supportive
    - palliative

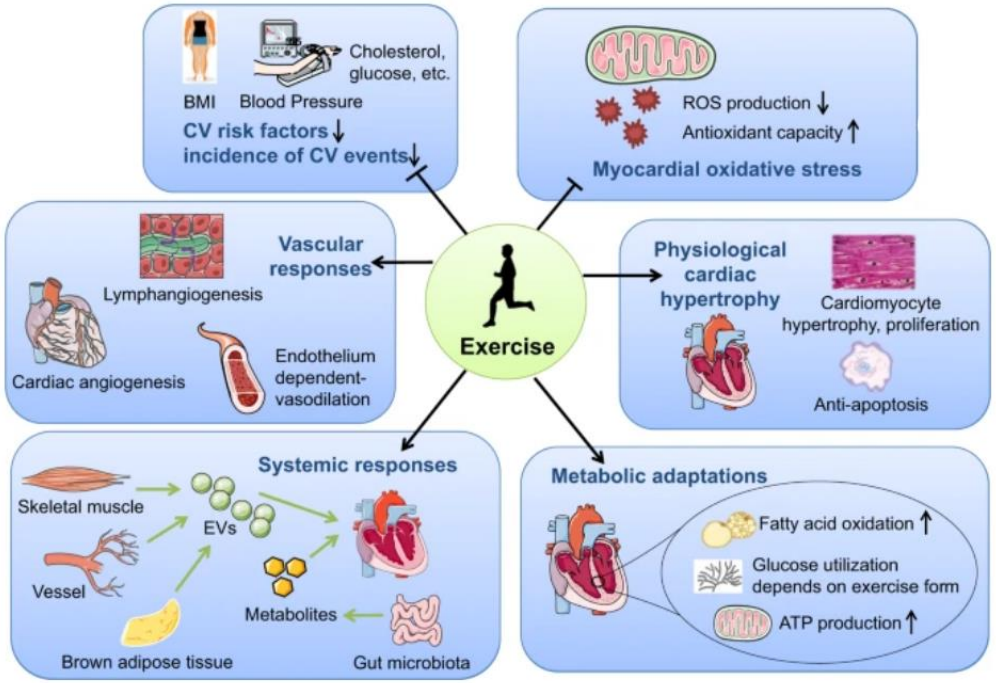
# How about fitness?

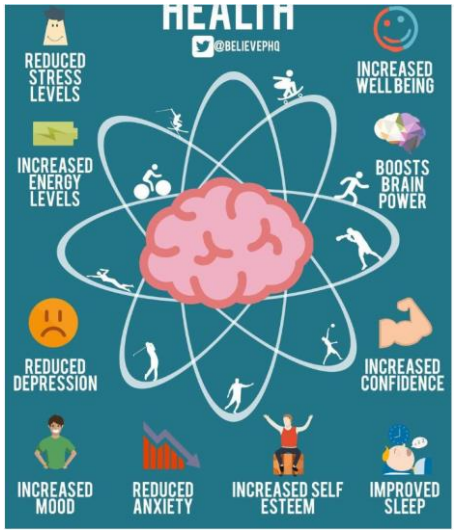
- A physiologic state of well-being that allows one to meet the demands of daily living and provides the basis for sports (task) performance or both

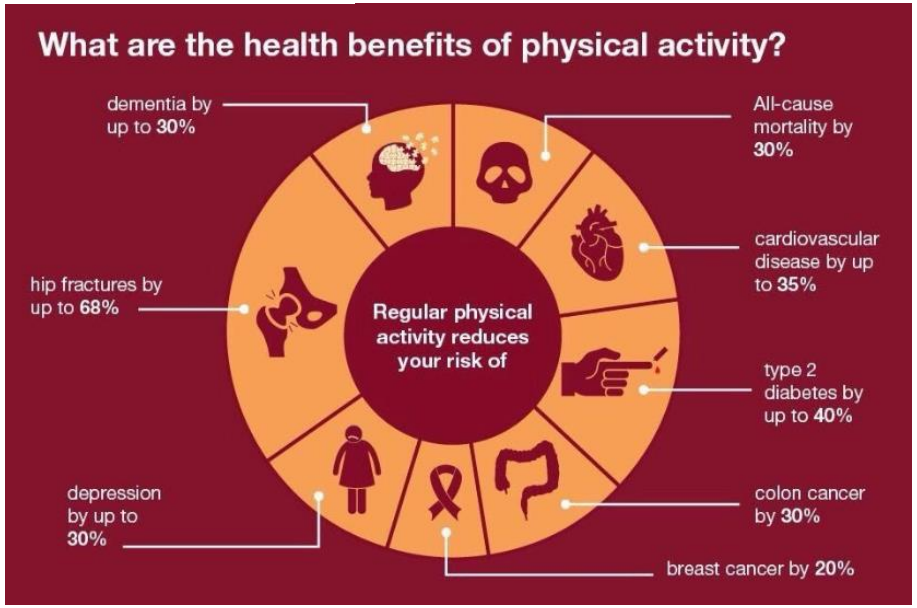


# What are the health benefits of physical activity?

Fig. 1



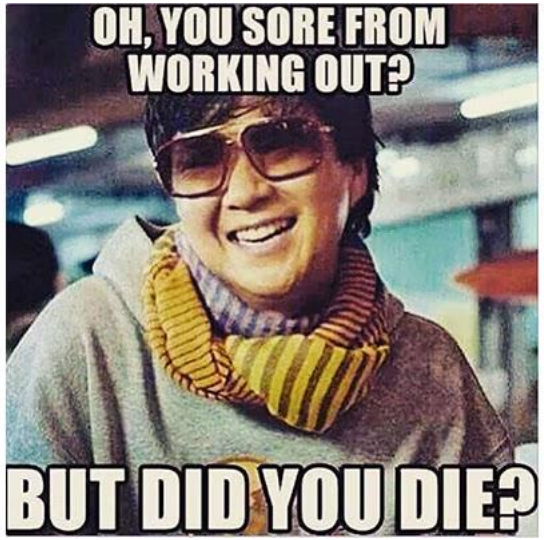
# Why is it important in cardiorespiratory patients in particular?

- High prevalence of comorbidity in people with respiratory disease that is influenceable with exercise

| | |
|---|---|
| Hypertension | 40% in COPD patients vs 27% in all adults |
| Skeletal muscle dysfunction | 10-15% loss of fat free mass in mild-mod COPD, up to 50% in severe |
| Osteoporosis | 50% in COPD patients; 30% in women / 20% in men |
| psychological disturbance | 42% in COPD |
| Disordered sleep | 10% |
| T2DM | 12% vs 9% in normal |
| Renal insufficiency | 22% vs 10% in general population |
| GORD | 60% vs 20% in general population of high income countries |

# Why is it important in cardiorespiratory patients in particular?

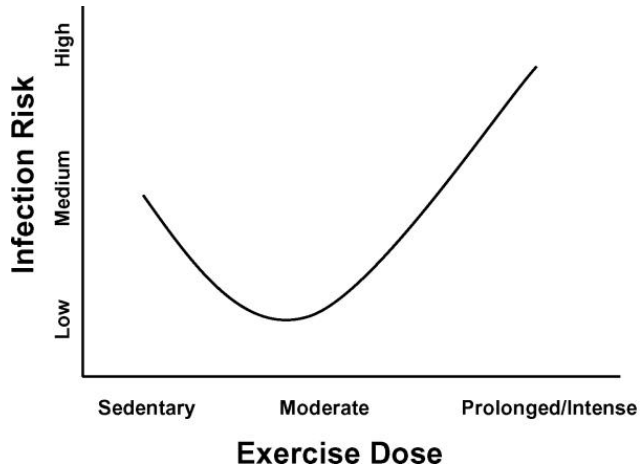
- Help to offset the complications associated with respiratory medications
  - Oral corticosteroids
    - Deranged blood sugars
    - Reduced bone density
    - Mood changes
    - Increased appetite / weight gain
    - Difficulty sleeping
    - GORD
    - Increased risk of fungal infections
    - Myopathy
  - Long term antibiotics
    - Altered immune response
  - Steroid inhaler
    - Risk of oral thrush
  - Antifibrotics
    - Diarrhoea
  - Mucolytics
    - Nausea
    - Weight loss

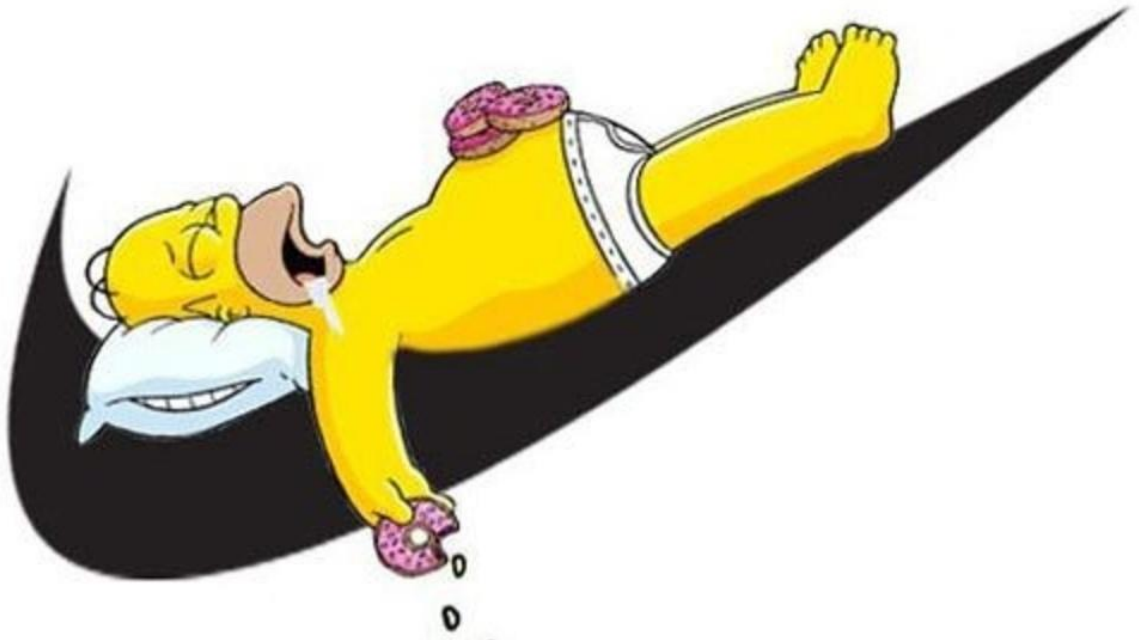


# Why is it important in cardiorespiratory patients in particular?

- Modulates elements of the disease itself

- Evidences support that exercise is able to modulate inflammatory pathways in asthma and COPD, including cytokines and leukotriene pathways
- Reversal/slow down cachexia in interstitial lung disease
- Strengthens immune response therefore reduce infection risks

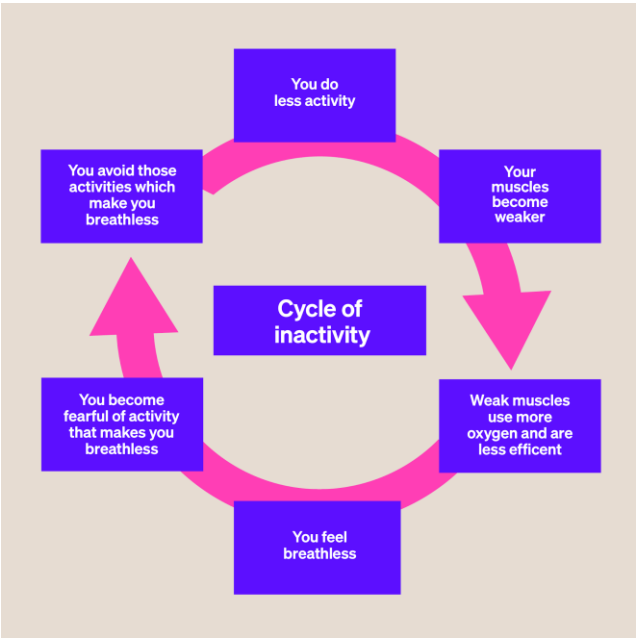




CAN'T SOMEONE ELSE JUST DO IT?

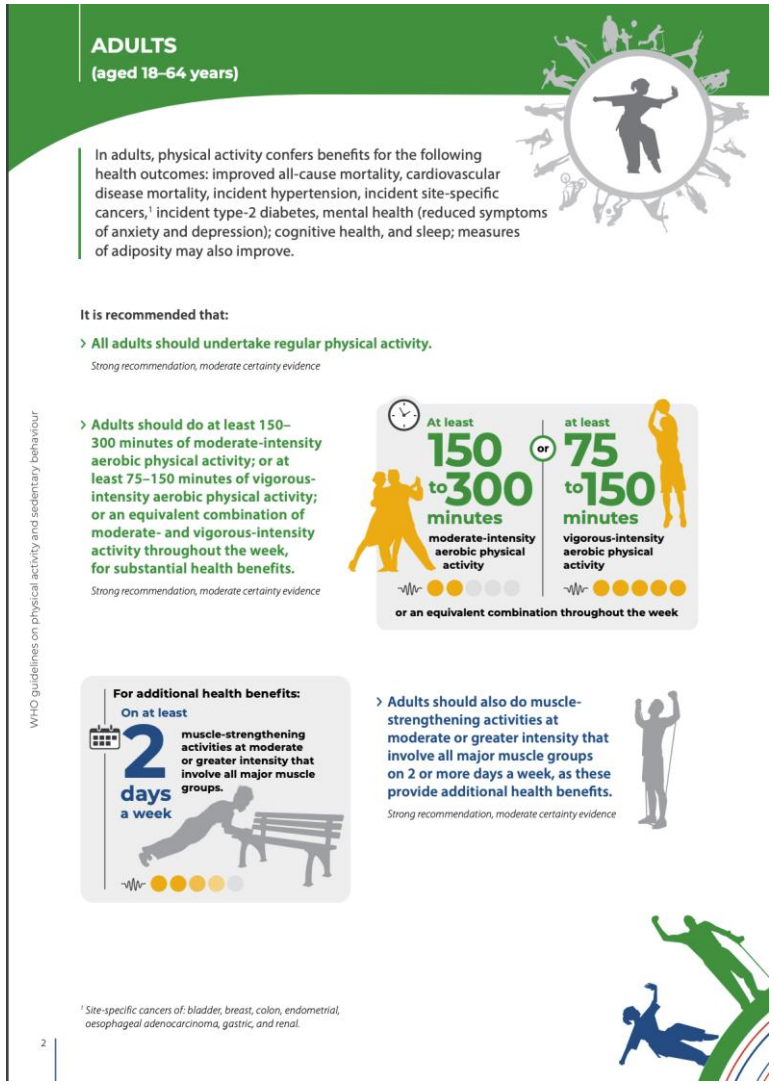
# What if you don't maintain physically active?

- Deconditioning is the decline in physical function of the body as a result of physical inactivity.
  - Atrophy of muscle bulks
  - Change in fibre types to more type 2 fast twitch
  - Reduced capillaries density
  - Worsening cardiac output
  - Reduced peripheral vascular tone
  - Stiffer and more effortful movement
  - Reduced bone density
  - Increase risk of injury




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# Current exercise guidelines



## ADULTS
(aged 18–64 years)

In adults, physical activity confers benefits for the following health outcomes: improved all-cause mortality, cardiovascular disease mortality, incident hypertension, incident site-specific cancers,[1] incident type-2 diabetes, mental health (reduced symptoms of anxiety and depression); cognitive health, and sleep; measures of adiposity may also improve.

**It is recommended that:**

› **All adults should undertake regular physical activity.**

*Strong recommendation, moderate certainty evidence*

› **Adults should do at least 150–300 minutes of moderate-intensity aerobic physical activity; or at least 75–150 minutes of vigorous-intensity aerobic physical activity; or an equivalent combination of moderate- and vigorous-intensity activity throughout the week, for substantial health benefits.**

*Strong recommendation, moderate certainty evidence*

At least **150 to 300 minutes** moderate-intensity aerobic physical activity

or

at least **75 to 150 minutes** vigorous-intensity aerobic physical activity

or an equivalent combination throughout the week

**For additional health benefits:**

On at least **2 days a week** muscle-strengthening activities at moderate or greater intensity that involve all major muscle groups.

› **Adults should also do muscle-strengthening activities at moderate or greater intensity that involve all major muscle groups on 2 or more days a week, as these provide additional health benefits.**

*Strong recommendation, moderate certainty evidence*

[1] *Site-specific cancers of: bladder, breast, colon, endometrial, oesophageal adenocarcinoma, gastric, and renal.*







**For additional health benefits:**

more than **300 minutes** moderate-intensity aerobic physical activity

or

more than **150 minutes** vigorous-intensity aerobic physical activity

or an equivalent combination throughout the week

› **Adults may increase moderate-intensity aerobic physical activity to more than 300 minutes; or do more than 150 minutes of vigorous-intensity aerobic physical activity; or an equivalent combination of moderate- and vigorous-intensity activity throughout the week for additional health benefits.**

*Conditional recommendation, moderate certainty evidence*

**GOOD PRACTICE STATEMENTS**

- Doing some physical activity is better than doing none.
- If adults are not meeting these recommendations, doing some physical activity will benefit their health.
- Adults should start by doing small amounts of physical activity, and gradually increase the frequency, intensity and duration over time.

In adults, higher amounts of sedentary behaviour are associated with the following poor health outcomes: all-cause mortality, cardiovascular disease mortality and cancer mortality and incidence of cardiovascular disease, cancer and type-2 diabetes.

**It is recommended that:**

› **Adults should limit the amount of time spent being sedentary. Replacing sedentary time with physical activity of any intensity (including light intensity) provides health benefits.**

*Strong recommendation, moderate certainty evidence*

› **To help reduce the detrimental effects of high levels of sedentary behaviour on health, adults should aim to do more than the recommended levels of moderate- to vigorous-intensity physical activity.**

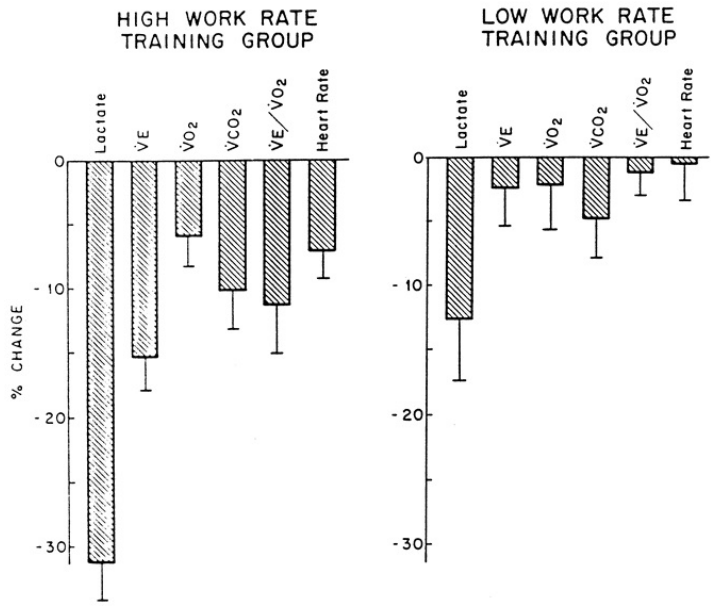
*Strong recommendation, moderate certainty evidence*

**LIMIT** the amount of time spent being sedentary

**REPLACE** with more physical activity of any intensity (including light intensity).

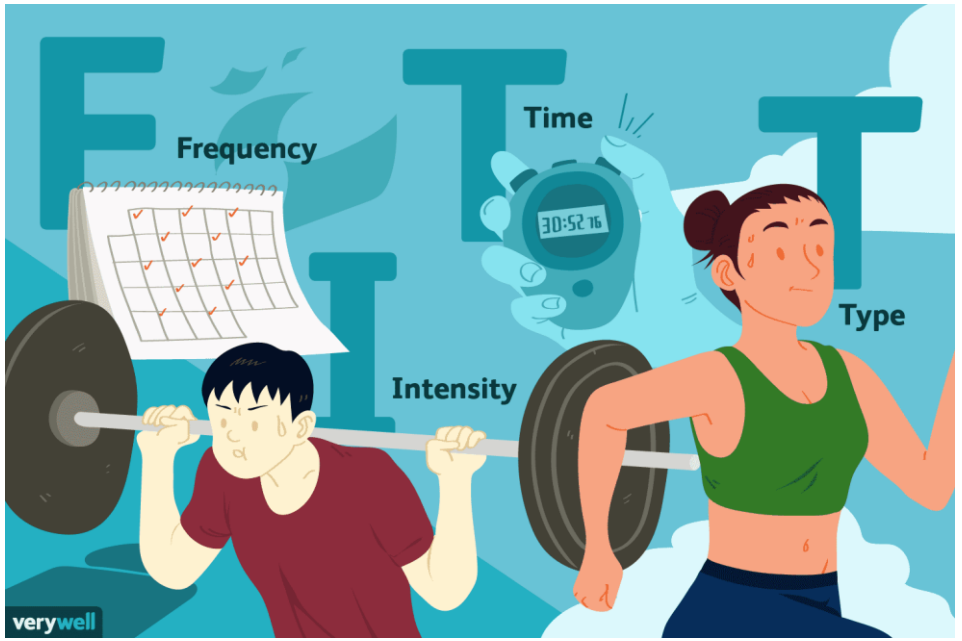
# Current exercise guideline

- What is the evidence behind the exercise guidelines
  - Frequency: 30 mins for 3 days a week increases 6MWT outcome by 66m
  - Time: >7 weeks showed improved outcome in dyspnea, emotion and mastery, >12 weeks showed improved health status and function
  - Intensity:

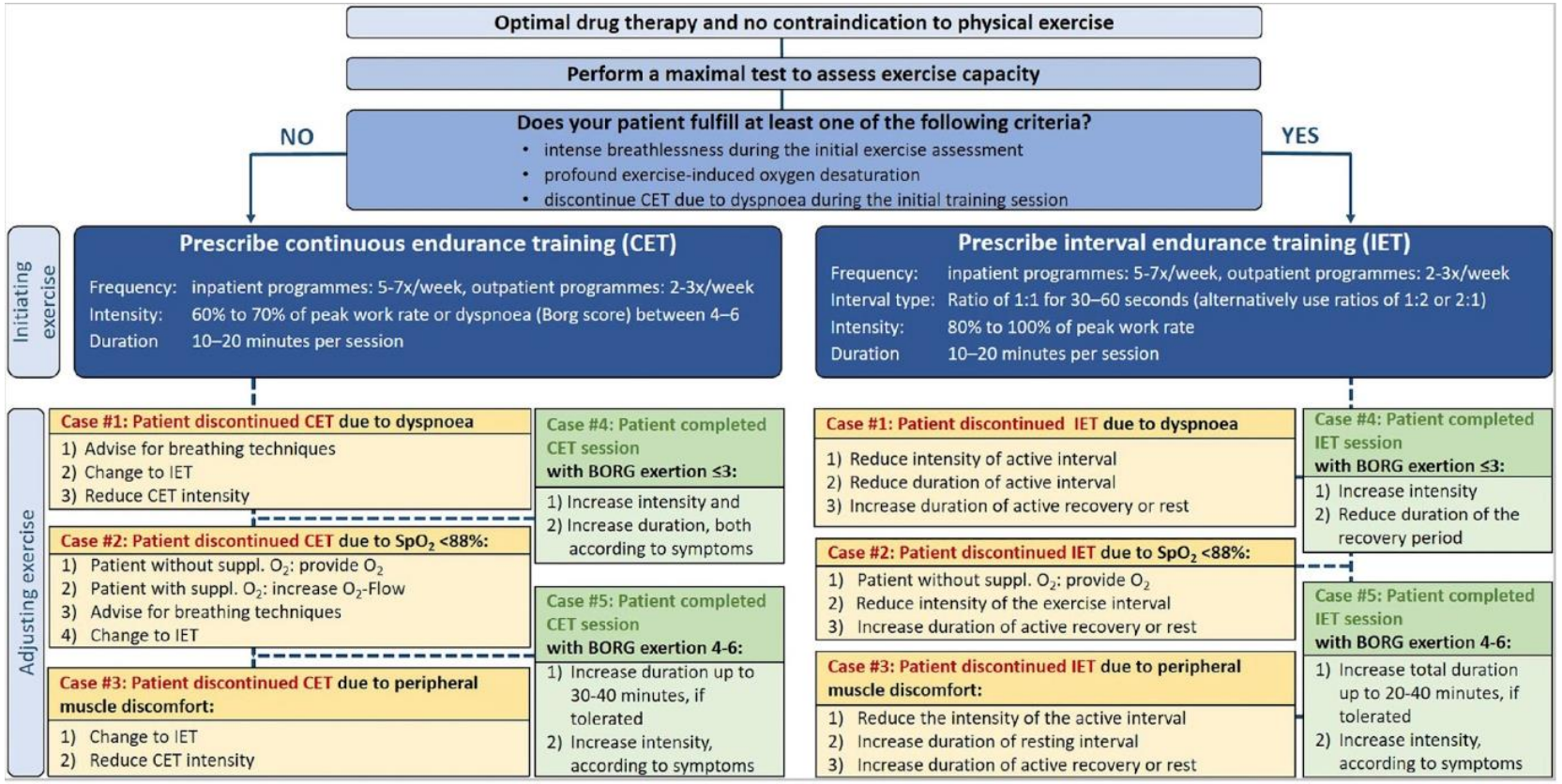


# What to prescribe to your patients

- FITT principle
  - **F**requency
  - **I**ntensity
  - **T**ime
  - **T**ype

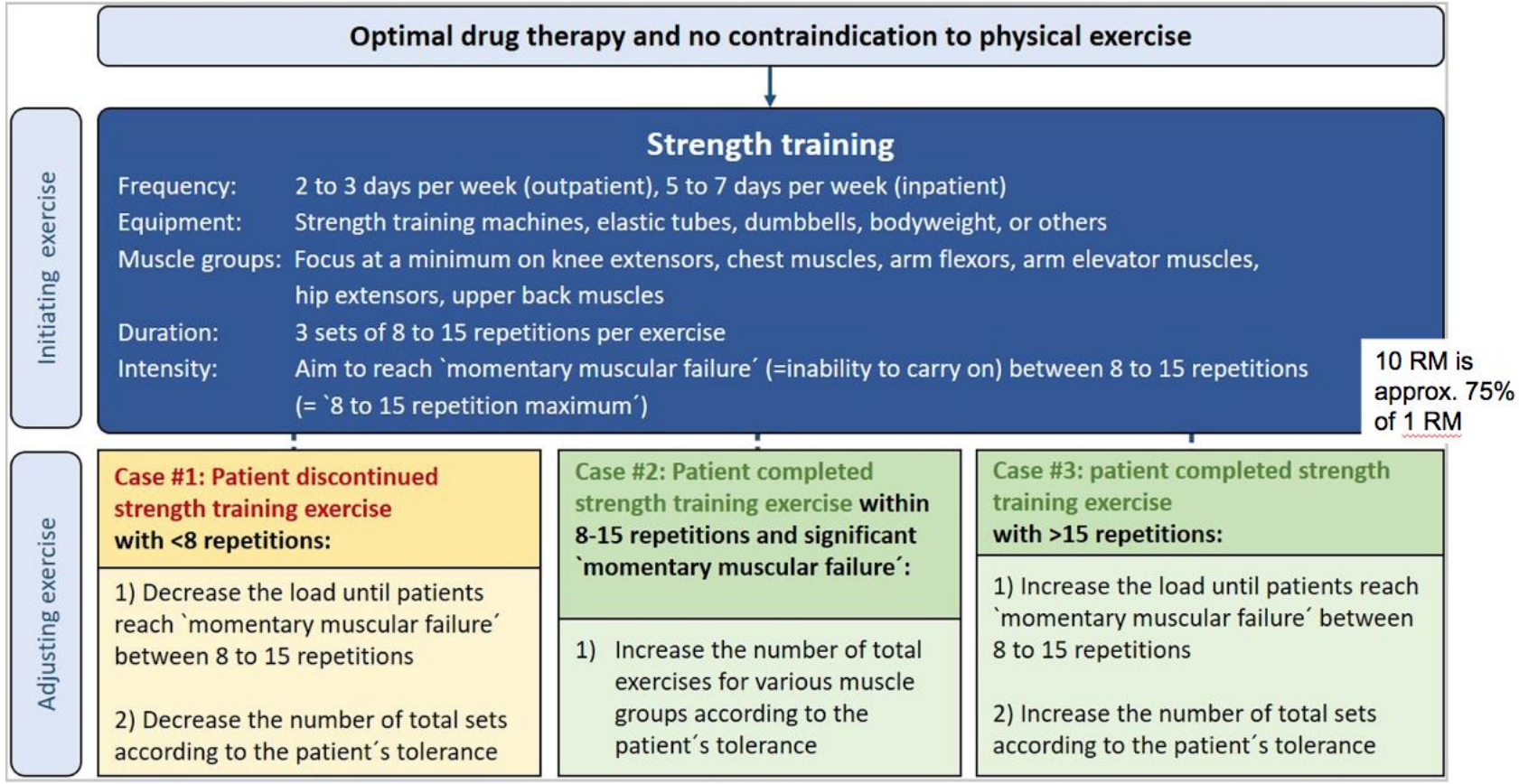


# What to prescribe to your patients




Gloeckl et al (2023) – Prescribing and adjusting ex training in Chronic Respiratory Disease

# What to prescribe to your patients



# What to prescribe to your patients

- Pacing progression
  - Want to avoid boom and bust
  - Avoid discouraging- Too much too soon
  - Be wary of the patient group- Not all will have exercised before
  - We want exercise to be a habit for life not a quick fix
  - Avoidance of post exertional malaise

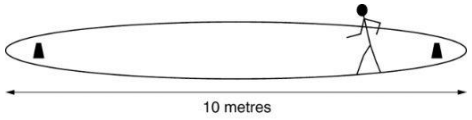



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# How do we test and measure exercise

- Cardiopulmonary Exercise Testing
  - Gold standard exercise testing
  - Able to provide comprehensive information on cardiac response, perfusion, respiratory
  - Bruce protocol
  - Can be completed on treadmill, bike or even other instruments
  - Can complete with other investigations in conjunction e.g laryngoscopy
- Incremental shuttle walk test / Beep test
- 6 minute walk test
  - Commonly used in oxygen prescription
  - Chronic respiratory disease monitoring
  - Transplant assessment
  - Has a learning effect
- 1 minute sit-to-stand test
  - Easy to complete
  - Good use to monitor trend
  - Less time consuming






The incremental shuttle walk test protocol

| Level | Speed (m/s) | Number of shuttles per level | Distance ambulated at the end of each level (m) |
|---|---|---|---|
| 1 | 0.50 | 3 | 30 |
| 2 | 0.67 | 4 | 70 |
| 3 | 0.84 | 5 | 120 |
| 4 | 1.01 | 6 | 180 |
| 5 | 1.18 | 7 | 250 |
| 6 | 1.35 | 8 | 330 |
| 7 | 1.52 | 9 | 420 |
| 8 | 1.69 | 10 | 520 |
| 9 | 1.86 | 11 | 630 |
| 10 | 2.03 | 12 | 750 |
| 11 | 2.20 | 13 | 880 |
| 12 | 2.37 | 14 | 1020 |

CLINICAL SCIENCES

**Validation and Cardiorespiratory Response of the 1-Min Sit-to-Stand Test in Interstitial Lung Disease**

PIERRE-FRANÇOIS TREMBLAY LABRECQUE, JANY HARVEY, ÉRIC NADREAU, FRANÇOIS MALTAIS, GENEVIÈVE DION, and DIDIER SAEY

*Centre de Recherche, Institut Universitaire de Cardiologie et de Pneumologie de Québec, Université Laval, Québec, CANADA*

# How do we test and measure exercise

- Calculation of maximal heart rate
  - 220 – age
  - Easy to calculate
  - Inaccurate by 7 to 11 bpm
  - Overestimates in under 30s, underestimates in over 45s
  - Varies between fitness level and type of activities

**Revisiting age-predicted maximal heart rate: Can it be used as a valid measure of effort?**

**Ross Arena, PhD, PT, FAHA**[a], **Jonathan Myers, PhD, FAHA**[b], and **Leonard A. Kaminsky, PhD**[c]

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[b]VA Palo Alto Health Care System and Stanford University, Ball State University, Muncie, IN

[c]Clinical Exercise Physiology, Ball State University, Muncie, IN

**Abstract**

**Introduction—**Despite high error ranges, age-predicted maximal heart rate (APMHR) is frequently used to gauge the achievement of adequate effort during an exercise test. The current analysis revisits this issue using the Fitness Registry and the Importance of Exercise: National Database (FRIEND Registry).

**Methods—**A total of 4,796 (63% male) apparently healthy subjects underwent a maximal cardiopulmonary exercise test on a treadmill. The mean age, maximal heart rate (HR), and

# How do we test and measure exercise

- Rate of Perceived Exertion (RPE)
  - Patient reported scale on effort of exertion
  - Costless and easy to use
  - Can be used by patient independently to pace physical activities
- Newly developed Dalhousie Dyspnoea and Perceived Exertion Scale

Does rating of perceived exertion result in target exercise intensity during interval training in cardiac rehabilitation? A study of the Borg scale versus a heart rate monitor

Inger-Lise Aamot [a b], Siv Hege Forbord [b], Trine Karlsen [a c], Asbjørn Støylen [c d]

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Abstract

Objectives

**A** **Modified Borg scale – Burdon et al[22]**

| | |
|---|---|
| 0 | Nothing at all |
| 0.5 | Very, very slight (just noticeable) |
| 1 | Very slight |
| 2 | Slight |
| 3 | Moderate |
| 4 | Somewhat severe |
| 5 | Severe |
| 6 | |
| 7 | Very severe |
| 8 | |
| 9 | Very, very severe (almost maximal) |
| 10 | Maximal |

| Rating | Descriptor |
|---|---|
| 6 | No exertion at all |
| 7 | Extremely light |
| 8 | |
| 9 | Very light |
| 10 | |
| 11 | Light |
| 12 | |
| 13 | Somewhat hard |
| 14 | |
| 15 | Hard (heavy) |
| 16 | |
| 17 | Very Hard |
| 18 | |
| 19 | Extremely hard |

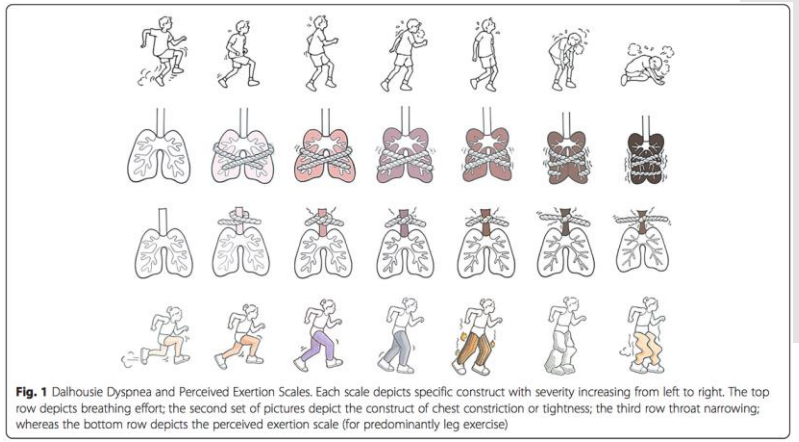


**Fig. 1** Dalhousie Dyspnea and Perceived Exertion Scales. Each scale depicts specific construct with severity increasing from left to right. The top row depicts breathing effort; the second set of pictures depict the construct of chest constriction or tightness; the third row throat narrowing; whereas the bottom row depicts the perceived exertion scale (for predominantly leg exercise)

# Exercise programmes specific to cardiorespiratory health

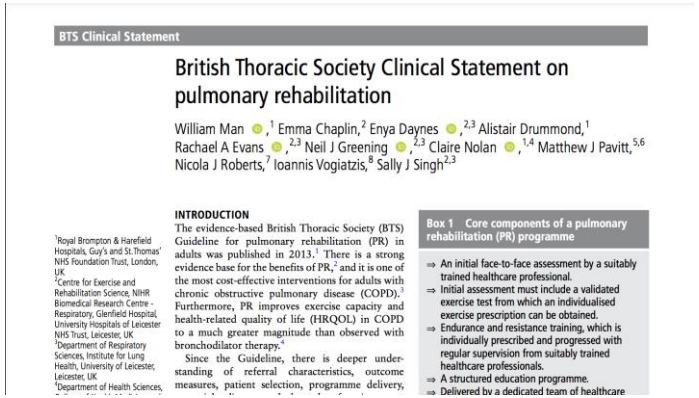
- **Pulmonary rehabilitation**
- Cardiac rehabilitation
- Arrhythmia exercise programme
- Postural Orthostatic Tachycardia rehabilitation
- Long COVID rehabilitation
- Breathing pattern disorder retraining

# Pulmonary rehabilitation

- A comprehensive intervention based on a thorough patient assessment followed by patient-tailored therapies that include, but are not limited to, exercise training, education, and behaviour change, designed to improve the physical and psychological condition of people with chronic respiratory disease and to promote the long-term adherence to health-enhancing behaviours

# Pulmonary rehabilitation

- Key intervention in GOLD guideline for COPD
  - One of the management with strongest evidence base
  - Latest research suggested early PR post discharge reduces re-admission rate by almost 50%
- Often as mandatory intervention prior to transplant
- Has own BTS guidelines and within NICE guidelines
- Mainly consist of COPD patient, but also extended to ILD and asthma patients in a lot of services



BTS Clinical Statement

British Thoracic Society Clinical Statement on pulmonary rehabilitation

William Man [1], Emma Chaplin [2], Enya Daynes [2,3], Alistair Drummond [1], Rachael A Evans [2,3], Neil J Greening [2,3], Claire Nolan [1,4], Matthew J Pavitt [5,6], Nicola J Roberts [7], Ioannis Vogiatzis [8], Sally J Singh [2,3]

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[4]Department of Health Sciences,

INTRODUCTION

The evidence-based British Thoracic Society (BTS) Guideline for pulmonary rehabilitation (PR) in adults was published in 2013.[1] There is a strong evidence base for the benefits of PR,[2] and it is one of the most cost-effective interventions for adults with chronic obstructive pulmonary disease (COPD).[3] Furthermore, PR improves exercise capacity and health-related quality of life (HRQOL) in COPD to a much greater magnitude than observed with bronchodilator therapy.[4]

Since the Guideline, there is deeper understanding of referral characteristics, outcome measures, patient selection, programme delivery,

Box 1 Core components of a pulmonary rehabilitation (PR) programme

⇒ An initial face-to-face assessment by a suitably trained healthcare professional.
⇒ Initial assessment must include a validated exercise test from which an individualised exercise prescription can be obtained.
⇒ Endurance and resistance training, which is individually prescribed and progressed with regular supervision from suitably trained healthcare professionals.
⇒ A structured education programme.
⇒ Delivered by a dedicated team of healthcare

# Pulmonary rehabilitation

- Consists of
  - 8-12 weeks of at least bi-weekly exercise sessions.
  - Mix of upper and lower limb exercises.
  - Mix of cardiovascular and strengthening exercises.
  - Educational component
- Hospital-based vs Virtual

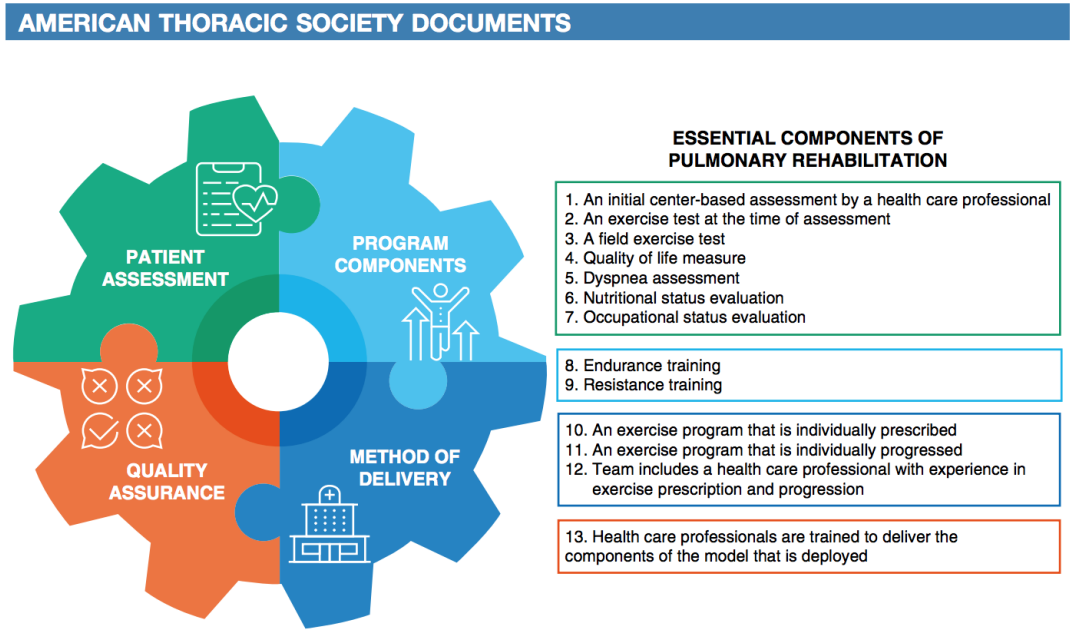



**Figure 1.** Essential components of pulmonary rehabilitation. Essential components of the pulmonary rehabilitation model were identified through a Delphi process. An essential component was defined as having a median score ≤2 (strongly agree or agree it is essential) and high consensus (interquartile range, 0).

# Pulmonary rehabilitation

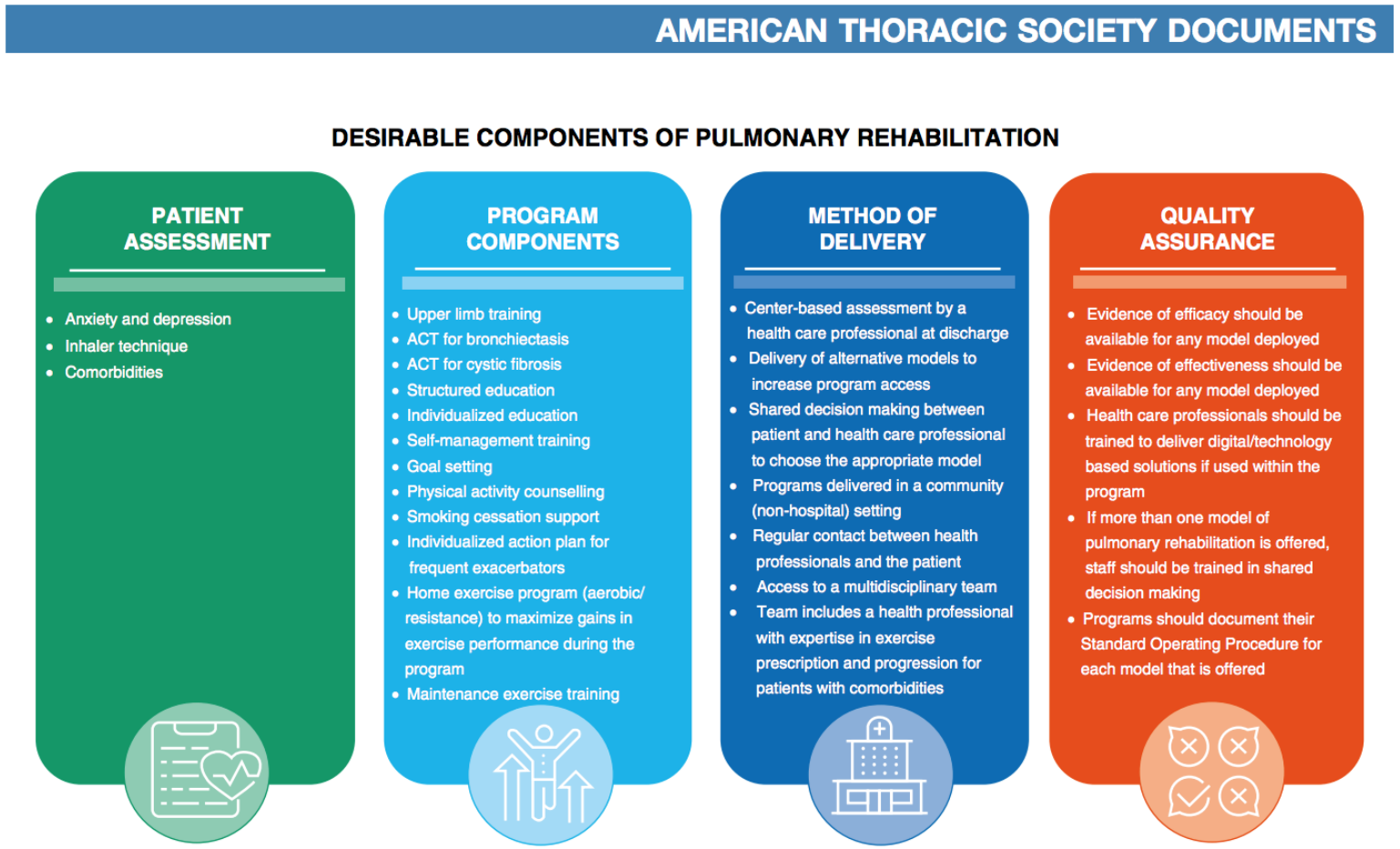
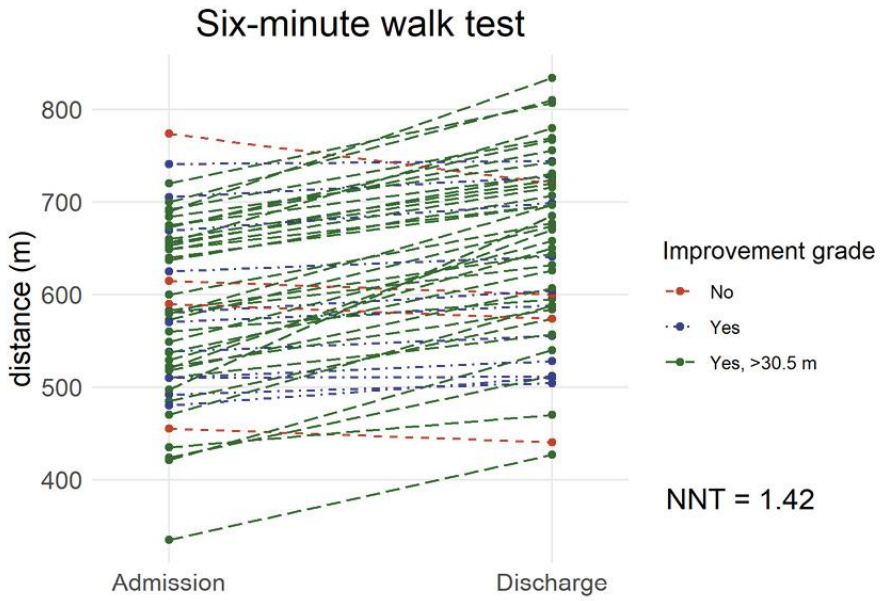
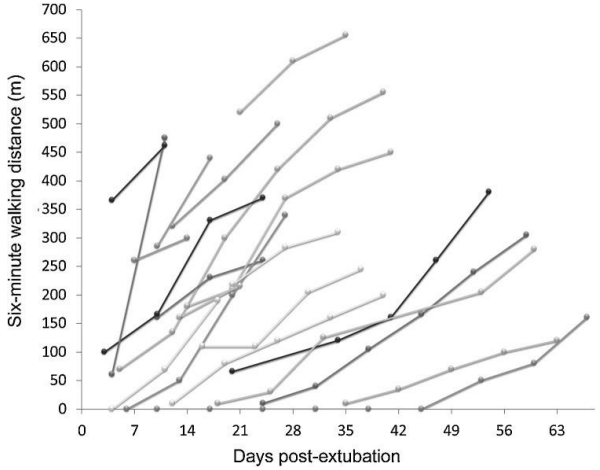
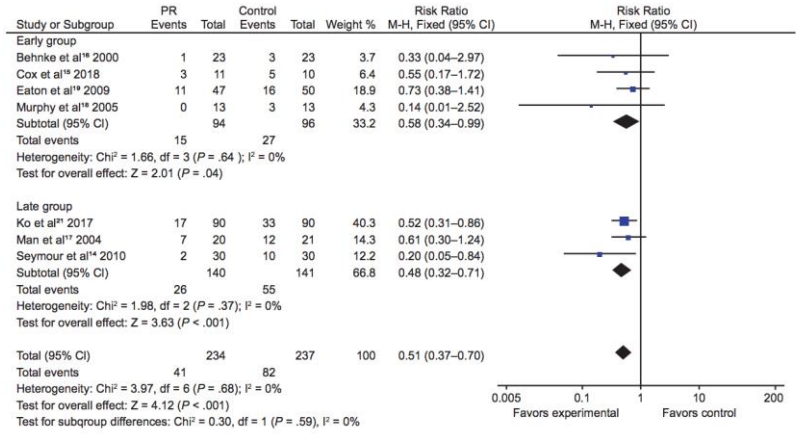



**Figure 2.** Desirable components of pulmonary rehabilitation. Desirable components of the pulmonary rehabilitation model were identified through a Delphi process. A desirable component was defined as having a median score ⩽2 (strongly agree or agree) but as having some variation in scores (interquartile range, 1). ACT = airway clearance techniques.

# Pulmonary Rehabilitation

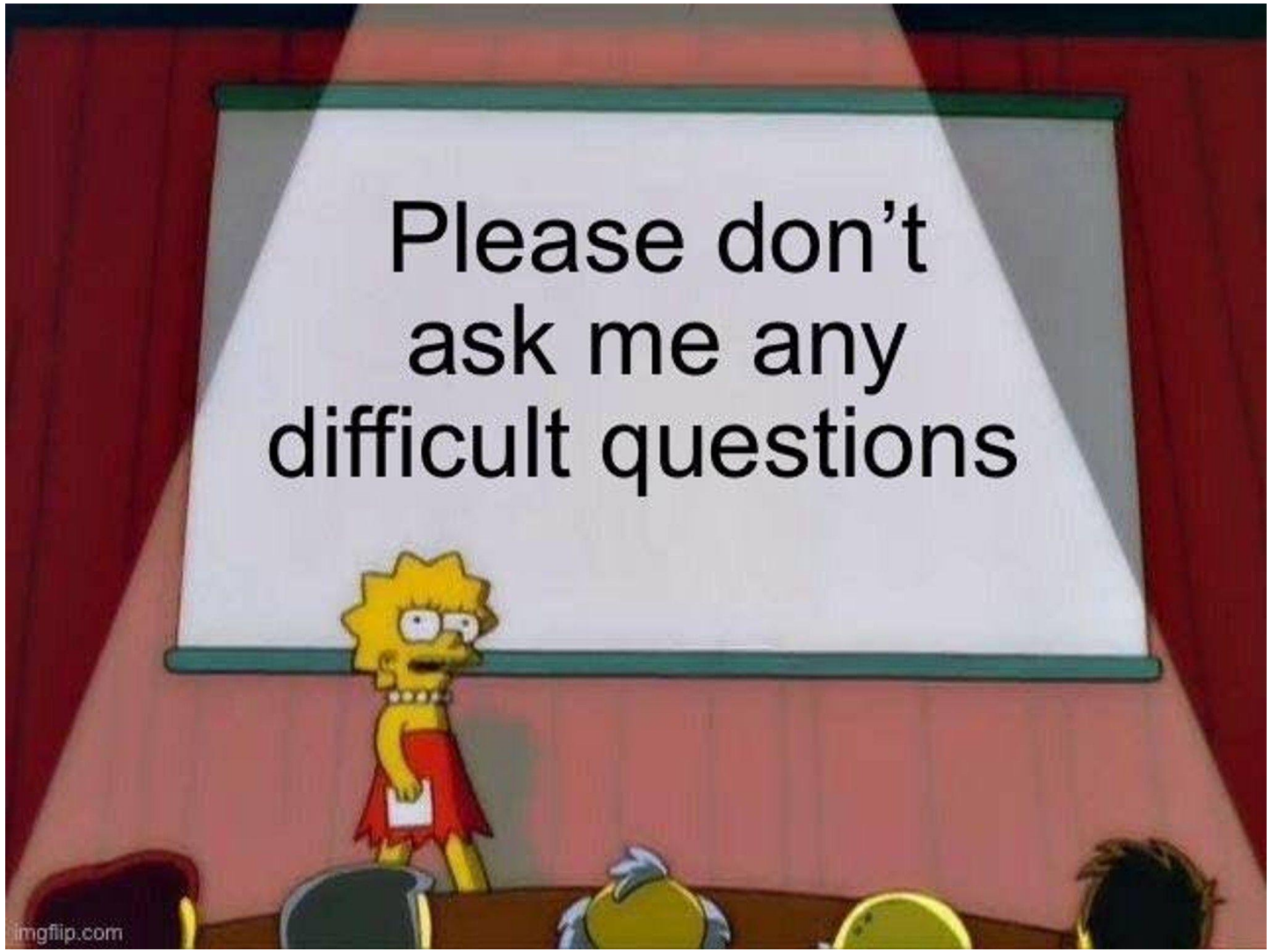



g. 2. The effect of pulmonary rehabilitation versus usual care on readmissions for up to 3–6 months. PR = pulmonary rehabilitation.

| | Within-Group Response to PR | | | |
|---|---|---|---|---|
| | IPF (n = 113) | | COPD (n = 103) | |
| Variable | Mean (95% CI) | *P* Value[a] | Mean (95% CI) | *P* Value[a] |
| ISWT distance change, m | 53 (37-69) | < .001 | 55 (44-66) | < .001 |
| MRC dyspnea scale grade change | −0.7 (−0.8 to −0.5) | < .001 | −0.7 (−0.9 to −0.6) | < .001 |
| CRQ score change | | | | |
| Dyspnea scale | 4.0 (2.9-5.1) | < .001 | 5.0 (3.7-6.2) | < .001 |
| Fatigue scale | 1.9 (1.0-2.8) | < .001 | 2.2 (1.3-3.1) | < .001 |
| Emotional function scale | 2.3 (1.0-3.5) | < .01 | 3.3 (2.0-4.7) | < .001 |
| Mastery scale | 1.4 (0.6-2.2) | < .001 | 2.2 (1.3-3.1) | < .001 |
| Total | 9.6 (6.5-12.6) | < .001 | 12.7 (9.2-16.2) | < .001 |


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